# HCIA-AI V4.0 Certification Practice Exam • Chapter 1

Module 1: AI Overview (100 Questions & Answer Key)

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## Question 1

TRUE OR FALSE • 16 Points

Speech processing is an AI discipline that only handles optical text character recognition, and cannot process acoustic voice signals or speech synthesis.

**A.** True
**B.** False

## Question 2

TRUE OR FALSE • 16 Points

Huawei's all-scenario AI solutions can be deployed in all kinds of environments, including public cloud, private clouds, various forms of edge computing, industrial IoT devices, and consumer devices.

**A.** True
**B.** False

## Question 3

TRUE OR FALSE • 16 Points

Artificial Intelligence (AI) is a subset of Machine Learning, and Machine Learning is a subset of Deep Learning.

**A.** True
**B.** False

## Question 4

TRUE OR FALSE • 16 Points

Foundation models require developers to train a completely new model from scratch with random initial weights for every distinct downstream task, completely prohibiting transfer learning.

**A.** True
**B.** False

## Question 5

TRUE OR FALSE • 16 Points

An AI smart speaker relies only on computer vision algorithms and does not use speech processing or natural language understanding technologies.

**A.** True
**B.** False

**Page Answers:** **Q1**: False • **Q2**: True • **Q3**: False • **Q4**: False • **Q5**: False

### Question 6

TRUE OR FALSE • 16 Points

Connectionism believes that intelligence originates from symbolic operations and knowledge representation systems rather than neural network connections.

**A.** True
**B.** False

### Question 7

TRUE OR FALSE • 16 Points

Behaviorism asserts that intelligence is formed through continuous interaction, perception, and action with the surrounding environment.

**A.** True
**B.** False

### Question 8

TRUE OR FALSE • 16 Points

The Turing Test was proposed by Alan Turing in 1950 to provide an operational definition of whether a machine can exhibit human-equivalent intelligence.

**A.** True
**B.** False

### Question 9

TRUE OR FALSE • 16 Points

The term 'Artificial Intelligence' was officially coined at the Dartmouth Conference held in the summer of 1956.

**A.** True
**B.** False

### Question 10

TRUE OR FALSE • 16 Points

Artificial General Intelligence (AGI) refers to narrow AI systems designed only to solve a single dedicated task, such as playing tic-tac-toe.

**A.** True
**B.** False

**Page Answers:** **Q6**: False • **Q7**: True • **Q8**: True • **Q9**: True • **Q10**: False

### Question 11

TRUE OR FALSE • 16 Points

In Natural Language Processing, Part-of-Speech (POS) tagging and Named Entity Recognition (NER) are typical sequence labeling tasks.

**A.** True
**B.** False

### Question 12

TRUE OR FALSE • 16 Points

Knowledge graphs represent structured real-world knowledge primarily in the form of entity-relation-entity triplets (head entity, relation, tail entity).

**A.** True
**B.** False

### Question 13

TRUE OR FALSE • 16 Points

Semantic segmentation classifies each pixel in an image into a category label, while distinguishing different individual instances of the exact same category.

**A.** True
**B.** False

### Question 14

TRUE OR FALSE • 16 Points

Optical Character Recognition (OCR) converts textual imagery captured by cameras or scanners into editable, machine-encoded text.

**A.** True
**B.** False

### Question 15

TRUE OR FALSE • 16 Points

The 'Emergence' phenomenon in large foundation models refers to complex capabilities (such as multi-step arithmetic or analogical reasoning) that appear spontaneously once the model parameter scale and data size surpass certain critical thresholds.

**A.** True
**B.** False

**Page Answers:** **Q11**: True • **Q12**: True • **Q13**: False • **Q14**: True • **Q15**: True

### Question 16

TRUE OR FALSE • 16 Points

DeepSeek models adopt Multi-Head Latent Attention (MLA) architecture primarily to significantly reduce KV cache memory consumption during inference while retaining performance.

**A.** True
**B.** False

### Question 17

TRUE OR FALSE • 16 Points

DeepSeek-R1 relies purely on supervised instruction fine-tuning without using reinforcement learning to achieve its complex mathematical and logical reasoning capabilities.

**A.** True
**B.** False

### Question 18

TRUE OR FALSE • 16 Points

In the Mixture of Experts (MoE) architecture used by advanced foundation models like DeepSeek-V3, all expert sub-networks are activated at the same time for every single token processed.

**A.** True
**B.** False

### Question 19

TRUE OR FALSE • 16 Points

Huawei's Pangu-Weather model uses 3D neural networks for medium-range global weather forecasting, achieving higher prediction speeds than conventional numerical weather prediction (NWP) systems.

**A.** True
**B.** False

### Question 20

TRUE OR FALSE • 16 Points

AI4Science refers to applying artificial intelligence techniques and large foundation models to fundamental scientific research fields such as drug discovery, materials science, and protein structure prediction.

**A.** True
**B.** False

**Page Answers:** **Q16**: True • **Q17**: False • **Q18**: False • **Q19**: True • **Q20**: True

### Question 21

TRUE OR FALSE • 16 Points

Watermarking generated content and establishing cryptographic provenance are technical countermeasures to address the spread of unauthorized deepfakes and AI-generated disinformation.

**A.** True
**B.** False

### Question 22

TRUE OR FALSE • 16 Points

Embodied AI focuses only on software chatbots living only in cloud servers with no connection to physical entities or robotic actuators.

**A.** True
**B.** False

### Question 23

TRUE OR FALSE • 16 Points

Data security and privacy protection laws (such as GDPR) require that personal data used in AI training must comply with principles of lawfulness, user consent, and data minimization.

**A.** True
**B.** False

### Question 24

TRUE OR FALSE • 16 Points

Chain of Thought (CoT) prompting strictly prohibits language models from generating intermediate reasoning steps and only allows direct one-word outputs.

**A.** True
**B.** False

### Question 25

TRUE OR FALSE • 16 Points

Model explainability (or interpretability) is of negligible importance when deploying AI in high-stakes domains like clinical medical diagnosis, autonomous driving, and judicial sentencing.

**A.** True
**B.** False

**Page Answers:** **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: False • **Q25**: False

### Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following correct hierarchical relationships describes Artificial Intelligence, Machine Learning, and Deep Learning?

- **A.** Deep Learning (subset of) AI (subset of) Machine Learning
- **B.** Machine Learning (subset of) Deep Learning (subset of) AI
- **C.** Deep Learning (subset of) Machine Learning (subset of) AI
- **D.** AI (subset of) Machine Learning (subset of) Deep Learning

### Question 27

SINGLE-ANSWER QUESTION • 16 Points

A manufacturer wants to produce virtual voice assistants for hospitals to handle patient inquiries and book appointments. Which of the following technical domains is primarily involved?

- **A.** Autonomous driving and SLAM navigation
- **B.** Computer vision and image reconstruction
- **C.** Natural language processing and speech recognition
- **D.** High-frequency RF antenna impedance tuning

### Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a typical use case for Natural Language Processing (NLP)?

- **A.** Text summarization of long legal documents
- **B.** Machine translation between English and French
- **C.** Sentiment analysis of product reviews
- **D.** Automatic image super-resolution upscaling

### Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following schools of artificial intelligence relies on simulated biological neural networks and connection weights to realize cognitive capabilities?

- **A.** Logicism
- **B.** Connectionism
- **C.** Behaviorism
- **D.** Symbolism

### Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI schools is rooted in cybernetics and evolutionary computing, arguing that intelligence develops through continuous interaction and feedback with the environment?

- **A.** Symbolism
- **B.** Structuralism
- **C.** Behaviorism
- **D.** Connectionism

**Page Answers:** **Q26**: C • **Q27**: C • **Q28**: D • **Q29**: B • **Q30**: C

### Question 31

SINGLE-ANSWER QUESTION • 16 Points

What are the three fundamental cornerstones (essential elements) driving the modern development of Artificial Intelligence?

- **A.** Python, PyTorch, and MindSpore
- **B.** Deep Learning, Reinforcement Learning, and Supervised Learning
- **C.** Data, Algorithms, and Computing Power
- **D.** Sensors, Internet, and Cloud Storage

### Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which Computer Vision task outputs both the bounding box coordinates and the semantic category label for each detected item in an image?

- **A.** Image Classification
- **B.** Object Detection
- **C.** Style Transfer
- **D.** Image Denoising

### Question 33

SINGLE-ANSWER QUESTION • 16 Points

In which historical event was the term 'Artificial Intelligence' first formally proposed?

- **A.** Dartmouth Summer Research Project on Artificial Intelligence
- **B.** The Turing Test Paper Publication
- **C.** The Victory of Deep Blue over Garry Kasparov
- **D.** The Invention of the Perceptron by Frank Rosenblatt

### Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which level of artificial intelligence possesses general cognitive abilities identical to or exceeding those of human beings across all intellectual domains?

- **A.** Artificial General Intelligence
- **B.** Expert Systems
- **C.** Artificial Narrow Intelligence
- **D.** Reactive Machines

### Question 35

SINGLE-ANSWER QUESTION • 16 Points

What is the primary architectural innovation introduced by DeepSeek to compress KV Cache storage requirements during large model inference?

- **A.** Grouped-Query Attention
- **B.** Multi-Head Latent Attention
- **C.** Multi-Head Attention
- **D.** Multi-Query Attention

**Page Answers:** **Q31**: C • **Q32**: B • **Q33**: A • **Q34**: A • **Q35**: B

### Question 36

SINGLE-ANSWER QUESTION • 16 Points

DeepSeek-V3 uses an MoE architecture. What does 'MoE' stand for in modern foundation models?

**A.** Matrix of Evaluation
**B.** Multi-objective Estimation
**C.** Model of Extraction
**D.** Mixture of Experts

### Question 37

SINGLE-ANSWER QUESTION • 16 Points

Which of the following NLP tasks classifies an entire sequence of words (e.g., an email) into predefined categories like 'Spam' or 'Ham'?

**A.** Sequence Labeling
**B.** Named Entity Recognition
**C.** Text Classification
**D.** Machine Translation

### Question 38

SINGLE-ANSWER QUESTION • 16 Points

A Knowledge Graph stores relationships between entities using triplets. Which of the following is the standard triplet format?

**A.** (Head Entity, Relation, Tail Entity)
**B.** (Primary Class, Feature Vector, Weight)
**C.** (Input Tensor, Activation, Output Bias)
**D.** (Source Node, Distance Metric, Target)

### Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the difference between Instance Segmentation and Semantic Segmentation?

**A.** Semantic segmentation labels each pixel by category, whereas instance segmentation additionally separates each individual object instance
**B.** Semantic segmentation isolates each individual instance separately, whereas instance segmentation aggregates all pixels into one object
**C.** Semantic segmentation is restricted to binary foreground masks, whereas instance segmentation performs holistic scene categorization
**D.** Semantic segmentation requires bounding box annotations, whereas instance segmentation relies only on image-level classification tags

### Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which of the following technologies enables an autonomous vehicle or surveillance camera to maintain continuous identity tracking of a pedestrian across sequential video frames?

**A.** Single Image Bounding Box Detection
**B.** Multiple Object Tracking Framework
**C.** Static Semantic Foreground Segmentation
**D.** Global Visual Scene Categorization

**Page Answers:** **Q36**: D • **Q37**: C • **Q38**: A • **Q39**: A • **Q40**: B

### Question 41

SINGLE-ANSWER QUESTION • 16 Points

In the Scaling Law of large foundation models, what primarily determines the predictable improvement in model cross-entropy loss?

- **A.** Optimizer momentum decay factor, dropout rate, and weight initialization seed
- **B.** Activation function non-linearity degree and batch normalization epsilon
- **C.** Gradient clipping threshold, learning rate warmup steps, and L2 penalty
- **D.** Number of model parameters, training dataset size, and total compute budget

### Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which prompting technique instructs a large language model to articulate intermediate reasoning steps to dramatically improve deductive problem-solving?

- **A.** Data Normalization and Z-Score Scaling
- **B.** Zero-Shot Chain of Thought
- **C.** Parameter-Efficient Low-Rank Adaptation
- **D.** Stochastic Gradient Descent with Momentum

### Question 43

SINGLE-ANSWER QUESTION • 16 Points

Huawei's AI-driven biodiversity monitoring initiative in Chile helped protect which endangered native species using acoustic pattern detection?

- **A.** African Elephant
- **B.** Giant Panda
- **C.** Polar Bear
- **D.** Darwin's fox

### Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei AI large model demonstrated remarkable precision in medium-range global meteorological forecasting, surpassing conventional Numerical Weather Prediction (NWP)?

- **A.** Pangu-Mine
- **B.** Pangu-Drug
- **C.** Pangu-Weather
- **D.** Pangu-Finance

### Question 45

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is considered a primary ethical challenge associated with generative AI and synthetic media?

- **A.** Transient electromagnetic interference in optical transceivers
- **B.** Exponential expansion of physical datacenter floor space limits
- **C.** Temporary increases in round-trip fiber network routing latency
- **D.** spread of fake videos, impersonation, and fraud

**Page Answers:** **Q41**: D • **Q42**: B • **Q43**: D • **Q44**: C • **Q45**: D

### Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI technologies combines perception, physical reasoning, and real-time robotic motor control to act in physical environments?

- **A.** Embodied AI
- **B.** Expert systems
- **C.** Linear regression
- **D.** K-means clustering

### Question 47

SINGLE-ANSWER QUESTION • 16 Points

What is the primary purpose of Automatic Speech Recognition (ASR) in an intelligent speech processing system?

- **A.** Compressing raw audio files without transcription
- **B.** Converting written text into synthesized audible human voice
- **C.** Converting spoken human acoustic audio signals into readable text
- **D.** Translating text directly between two different languages

### Question 48

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of Text-to-Speech (TTS) technology?

- **A.** Translating spoken audio from one dialect to another
- **B.** Filtering out background noise from recorded audio
- **C.** Extracting keywords from audio transcripts
- **D.** Synthesizing spoken audio output from written text input

### Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI pioneers proposed the famous concept of a 'Universal Computing Machine' and formulated the imitation game test for machine intelligence?

- **A.** Claude Shannon
- **B.** Alan Turing
- **C.** Norbert Wiener
- **D.** John von Neumann

### Question 50

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an AI for Science (AI4Science) application domain?

- **A.** Predicting 3D protein folding structures
- **B.** Developing fixed rule-based automation
- **C.** Discovering novel battery crystal materials
- **D.** Simulating high-energy molecular dynamics

**Page Answers:** **Q46**: A • **Q47**: C • **Q48**: D • **Q49**: B • **Q50**: B

### Question 51

SINGLE-ANSWER QUESTION • 16 Points

What does a 'Token' typically represent in the context of large language model processing?

- **A.** A complete raw document file stored in the training dataset
- **B.** The basic atomic semantic text unit parsed by a model
- **C.** A fixed-length sentence paragraph separated by newline characters
- **D.** A trainable weight matrix inside the feed-forward network layer

### Question 52

SINGLE-ANSWER QUESTION • 16 Points

What major open-source contribution did DeepSeek provide to the global AI development community?

- **A.** Mandating proprietary commercial licenses on all transformer attention weights
- **B.** Banning the deployment of mixture-of-experts architectures across private clouds
- **C.** Releasing open model weights and architectural reports achieving frontier performance
- **D.** Requiring static compilation of all dynamic computation graphs prior to execution

### Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which Computer Vision generative architecture uses a forward noise-adding process followed by a learned reverse denoising process to create high-fidelity images?

- **A.** Diffusion Models
- **B.** K-Nearest Neighbors
- **C.** Support Vector Regression
- **D.** Linear Discriminant Analysis

### Question 54

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an example of Narrow AI (Weak AI)?

- **A.** AlphaGo playing Go at superhuman levels
- **B.** A general robot capable of inventing scientific theories and raising human children seamlessly
- **C.** An omniscient superintelligence governing human civilization
- **D.** A conscious machine possessing autonomous subjective emotion and free will

### Question 55

SINGLE-ANSWER QUESTION • 16 Points

Which phase is commonly known as an 'AI Winter' in the history of artificial intelligence development?

- **A.** Memory bus interface parity check error interruptions
- **B.** Operating system kernel freezes caused by unhandled memory leaks during neural training runs
- **C.** Periods when rule-based expert systems completely solved general human intelligence
- **D.** Prolonged periods when funding, commercial interest, and academic research declined sharply

**Page Answers:** **Q51**: B • **Q52**: C • **Q53**: A • **Q54**: A • **Q55**: D

### Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Hallucination' problem in Large Language Models?

- **A.** The model generates plausible-sounding but factually false or unverifiable statements confidently
- **B.** The model encounters catastrophic memory overflow during tokenization of long-context windows
- **C.** The model produces fixed identical output embeddings regardless of the temperature parameter
- **D.** The model vocabulary lookup table fails to tokenize out-of-vocabulary mathematical operator symbols

### Question 57

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of 'Computing Power' (e.g., AI accelerators like Huawei Ascend NPUs) in modern AI?

- **A.** Generating high-precision ground truth labels automatically without programmatic human review
- **B.** Providing massive parallel matrix calculation throughput required to train and infer deep models
- **C.** Executing optical surface contour scanning for geometric solid modeling
- **D.** Formulating core loss function equations mathematically without requiring human intervention

### Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Natural Language Processing, which task aims to identify boundaries of real-world entities (such as person names, organization names, and geographic locations)?

- **A.** Named Entity Recognition
- **B.** Part-of-Speech Tagging
- **C.** Word Stemming
- **D.** Machine Translation

### Question 59

SINGLE-ANSWER QUESTION • 16 Points

What does the Dartmouth Workshop in 1956 signify in computer science history?

- **A.** The creation of the Linux operating system kernel
- **B.** The formulation of the World Wide Web by Tim Berners-Lee
- **C.** The invention of the world's first electronic computer
- **D.** The birth of Artificial Intelligence as a distinct academic discipline

### Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding reasoning models like DeepSeek-R1 are TRUE? (Multiple Choice)

- **A.** They employ large-scale reinforcement learning to develop verifiable reasoning capabilities
- **B.** They autonomously generate and verify intermediate step-by-step chain-of-thought deductions
- **C.** They require supervised fine-tuning on millions of human-annotated reasoning trajectories
- **D.** They rely on static non-contextual n-gram frequency lookup tables without self-attention

**Page Answers:** **Q56**: A • **Q57**: B • **Q58**: A • **Q59**: D • **Q60**: AB

### Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized AI ethical concerns widely debated internationally?

- **A.** Algorithmic bias and discriminatory outcomes in automated scoring systems
- **B.** Low-frequency audible vibration produced by server power transformer coils
- **C.** spread of fake videos, synthetic impersonation, and automated fraud
- **D.** Data privacy violations from unauthorized scraping and mass biometric collection

### Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Weak AI (ANI) and Strong AI (AGI) are correct?

- **A.** Strong AI has already been completely realized and deployed across enterprise cloud platforms
- **B.** Weak AI is designed and optimized to excel at specific, predefined narrow cognitive tasks
- **C.** All currently deployed commercial AI applications belong to the category of Weak AI
- **D.** Weak AI algorithms cannot be executed on modern multi-core GPU computing clusters

### Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In enterprise intelligence, which of the following represent typical deployment scenarios of Artificial Intelligence?

- **A.** Unmanned haulage trucks and visual conveyor belt inspection in Intelligent Mining
- **B.** Automated optical inspection for defect detection in Smart Manufacturing
- **C.** Replacing relational database transaction logs with unindexed raw text files
- **D.** Dynamic grid power dispatching and equipment fault prediction in Intelligent Energy

### Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In the AI industry ecosystem, which of the following components constitute the foundational Infrastructure and Framework layers?

- **A.** Specialized AI processors, NPUs, GPUs, and high-performance server clusters
- **B.** End-user domain applications, smart city solutions, and industry SaaS workflows
- **C.** Deep learning frameworks, operator acceleration libraries, and compiler toolchains
- **D.** Business process outsourcing, customer relationship management, and ERP portals

### Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following generative neural network architectures and models are widely applied for synthetic data and media generation?

- **A.** Diffusion Models
- **B.** Generative Adversarial Networks
- **C.** Variational Autoencoders
- **D.** Autoregressive Transformer Models

**Page Answers:** **Q61**: ACD • **Q62**: BC • **Q63**: ABD • **Q64**: AC • **Q65**: ABCD

### Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following AI schools of thought are primarily grounded in bionics and environmental interaction respectively?

- **A.** Symbolism
- **B.** Connectionism
- **C.** Behaviorism
- **D.** Functionalism

### Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major subfields and application domains of Artificial Intelligence?

- **A.** Speech Processing
- **B.** Knowledge Graph
- **C.** Natural Language Processing
- **D.** Computer Vision

### Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational Computer Vision (CV) visual perception tasks?

- **A.** Semantic and Instance Segmentation
- **B.** 2D and 3D Object Detection
- **C.** Syntactic Dependency Tree Parsing and Part-of-Speech Tagging
- **D.** Image Classification and Optical Character Recognition

### Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Natural Language Processing (NLP) is primarily divided into which core functional branches?

- **A.** Natural Language Understanding
- **B.** Speech Synthesis
- **C.** Natural Language Generation
- **D.** Image Classification

### Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core characteristics and advantages of large foundation models compared to traditional small models?

- **A.** Exact identical output reproduction across arbitrary random temperature values
- **B.** Emergent abilities at large parameter and data scales
- **C.** Stronger in-context learning and zero-shot reasoning capabilities
- **D.** Strong generalization across diverse downstream tasks through pre-training

**Page Answers:** **Q66**: BC • **Q67**: ABCD • **Q68**: ABD • **Q69**: AC • **Q70**: BCD

### Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core architectural innovations introduced in DeepSeek-V3 to optimize inference KV cache and sparse computation?

- **A.** Multi-Head Latent Attention
- **B.** Dense Convolutional Residual Stacking
- **C.** DeepSeekMoE architecture with shared experts
- **D.** Static Global Average Pooling

### Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors contributed to the rapid rise and global popularity of DeepSeek?

- **A.** Open availability of model weights and full technical transparency
- **B.** Outstanding performance competitive with top proprietary frontier models
- **C.** Mandatory subscription fees exceeding $10,000 per user
- **D.** Greatly reduced training and inference costs

### Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are landmark achievements of AI for Science (AI4Science) in biology and meteorology respectively?

- **A.** AlphaFold for 3D protein structure prediction
- **B.** Word2Vec for static lexical word embedding generation
- **C.** Pangu-Weather for global numerical weather forecasting
- **D.** AlexNet for 2D natural image category classification

### Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are ethical, social, and legal issues raised by the rapid growth of AI?

- **A.** Privacy leakage and unauthorized profiling of individuals
- **B.** Algorithmic bias resulting from unrepresentative training data
- **C.** Potential disruption of traditional employment and labor markets
- **D.** Copyright infringement and intellectual property ownership of generated works

### Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

In a Knowledge Graph, a basic semantic assertion is stored as a 'triplet'. which types of primary components constitute the nodes and edges of this triplet?

- **A.** Entities
- **B.** Biases
- **C.** Relations
- **D.** Gradients

**Page Answers:** **Q71**: AC • **Q72**: ABD • **Q73**: AC • **Q74**: ABCD • **Q75**: AC

### Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the two main tasks of speech processing?

- **A.** Image Segmentation
- **B.** Speech Recognition
- **C.** Object Detection
- **D.** Speech Synthesis

### Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are characteristics of Artificial Narrow Intelligence (Weak AI)?

- **A.** Lack of genuine self-awareness and conscious intentionality
- **B.** Highly effective at solving specific, well-defined problems
- **C.** Universal human-like general cognitive adaptability
- **D.** Inability to autonomously transfer learned skills to unrelated cognitive domains

### Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

What types of data are typically used in the pre-training of multimodal foundation models?

- **A.** text datasets from books, encyclopedias, and web documents
- **B.** Video clips and temporal action sequences
- **C.** High-resolution image collections with text descriptions
- **D.** Audio recordings and speech transcripts

### Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following measures can reduce security and privacy risks when training and serving AI models?

- **A.** Data anonymization and de-identification
- **B.** Storing unencrypted raw biometric identifiers in public training repositories
- **C.** Differential privacy mechanisms during gradient updates
- **D.** Robust access control and model output content filtering

### Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are standard implementations of Chain of Thought (CoT) prompting in Large Language Models?

- **A.** Zero-Shot-CoT
- **B.** Weight Pruning
- **C.** Few-Shot-CoT
- **D.** Model Quantization

**Page Answers:** **Q76**: BD • **Q77**: ABD • **Q78**: ABCD • **Q79**: ACD • **Q80**: AC

### Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hardware and software layers comprise Huawei's full-stack AI solution?

- **A.** Ascend AI processor hardware series
- **B.** Compute Architecture for Neural Networks chip operator library
- **C.** ModelArts one-stop AI development platform
- **D.** MindSpore AI framework

### Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are examples of Generative AI applications in real life?

- **A.** Seismic waveform deconvolution and strata modeling
- **B.** Text-to-Image creation of artworks and product designs
- **C.** AI-assisted code generation and completion tools
- **D.** Automated essay drafting and conversational assistants

### Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the primary computing chip architectures powering modern high-performance AI model training and inference?

- **A.** Legacy Mechanical Relays
- **B.** Graphics Processing Units
- **C.** Magnetic Tape Storage Controllers
- **D.** Neural Processing Units

### Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are main features of the Behaviorism school of AI?

- **A.** Inspired by biological reflexes and adaptive behaviors
- **B.** Emphasis on perception-action loops with the environment
- **C.** Exclusive reliance on First-Order Predicate Logic theorem proving
- **D.** Foundation in reinforcement learning and evolutionary algorithms

### Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key milestones in the historical development of Artificial Intelligence?

- **A.** 2016: AlphaGo defeats world champion Lee Sedol
- **B.** 1950: Alan Turing proposes the Turing Test
- **C.** 1956: The Dartmouth Workshop coins the term AI
- **D.** 1997: IBM Deep Blue defeats world chess champion Garry Kasparov

**Page Answers:** **Q81**: ABCD • **Q82**: BCD • **Q83**: BD • **Q84**: ABD • **Q85**: ABCD

## Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major advantages of Mixture of Experts (MoE) models?

**A.** Specialized sub-networks focusing on distinct knowledge domains
**B.** Massive model parameter capacity without a proportional increase in FLOPs per token
**C.** Eliminating the need for training data completely
**D.** Lower inference computation latency compared to dense models of equivalent total parameter size

## Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized challenges or technical limitations of current Large Language Models?

**A.** Hallucination
**B.** Inability to execute basic string concatenation
**C.** Total absence of zero-shot in-context adaptation
**D.** Massive computational costs during pre-training and scaling

## Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Embodied AI are correct?

**A.** It requires close integration of perception, spatial reasoning, planning, and motor dynamics
**B.** Embodied AI agents perceive the physical world via multimodal sensors and act via physical bodies or manipulators
**C.** It is fundamental to the development of next-generation humanoid robotics
**D.** It operates strictly on text strings without any spatial or physical interaction

## Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods can be used to improve the security and fairness of AI systems?

**A.** Testing model safety and output quality before deployment
**B.** Deploying medical models without testing their accuracy
**C.** Making models explainable and understanding their decisions
**D.** Protecting data privacy and checking datasets for bias

## Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are downstream tasks enabled by pre-trained foundation models?

**A.** Multilingual document translation and summarization
**B.** Creative writing and programmatic code completion
**C.** Open-domain question answering and dialogue
**D.** Automated legal contract analysis

**Page Answers:** **Q86**: ABD • **Q87**: AD • **Q88**: ABC • **Q89**: ACD • **Q90**: ABCD

### Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized as the core driving elements (cornerstones) of modern Artificial Intelligence?

- **A.** High-quality Big Data
- **B.** Massive Computing Power
- **C.** Advanced Machine Learning and Deep Learning Algorithms
- **D.** First-Order Predicate Logic Theorem Proving

### Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following historical milestones are landmark events in the development of Artificial Intelligence?

- **A.** The 1956 Dartmouth Summer Research Project on Artificial Intelligence
- **B.** The 1997 Deep Blue victory over world chess champion Garry Kasparov
- **C.** The 2016 AlphaGo victory in the ancient board game of Go
- **D.** The 1943 McCulloch-Pitts formal mathematical artificial neuron model

### Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

Artificial intelligence that is designed to perform a single specific dedicated task (such as facial recognition or playing chess) is known as ____ AI.

**Answer:** _____

### Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In natural language processing, NER stands for Named Entity ____.

**Answer:** _____

### Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

Knowledge graphs represent structured relational knowledge primarily using ____, consisting of a head entity, a relation, and a tail entity.

**Answer:** _____

**Page Answers:** **Q91**: ABC • **Q92**: ABCD • **Q93**: Narrow (or Weak) • **Q94**: Recognition • **Q95**: triplets (or triples)

### Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

The computer vision technology used to recognize and digitize printed or handwritten characters from images into machine-readable text is called ____.

**Answer:** _____

### Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In modern large language models, CoT stands for Chain of ____.

**Answer:** _____

### Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

DeepSeek models use MLA, which stands for Multi-Head ____ Attention, to optimize KV cache memory during inference.

**Answer:** _____

### Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In MoE architecture, MoE stands for Mixture of ____.

**Answer:** _____

### Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's large meteorological AI model designed for high-accuracy global weather forecasting is called Pangu-____.

**Answer:** _____

**Page Answers:** **Q96**: OCR (or Optical Character Recognition) • **Q97**: Thought • **Q98**: Latent • **Q99**: Experts • **Q100**: Weather